Project Report

Title: Smart Expense Tracker

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Department: Electronics and Computer Engineering
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1.Project Overview

This project is a Python-based application called Smart Expense Tracker.
It is developed using Tkinter library for GUI. The main purpose of this project is to help users manage their daily expenses in a simple and organized way.

The system allows the user to add, delete, search, and view expenses. It also calculates the total expenses automatically and stores data in a CSV file.

2. Problem Statement

In daily life, people spend money on different things but often forget to keep track of it. This can lead to poor money management.

The aim of this project is to develop a simple and user-friendly system that helps users record and manage their expenses efficiently. The system should be easy to use and should store the data properly for future use.

3. Technology Stack

- Programming Language: Python 3.x
- IDE: Python IDLE

4. Implementation

The project is implemented using Python and Tkinter. The working of the system is as follows:

1. The program starts with a welcome screen.
2. User enters the dashboard by clicking "Enter Dashboard".
3. User can input:
    - Expense Name
    - Amount
    - Category
    - Date
4. Data is validated before adding.
5. Expenses are displayed in a table format.
6. User can:
    - Add expense
    - Delete selected expense
    - Search expense
    - Clear input fields
7. Total expense is calculated automatically.
8. All data is saved in a CSV file.

5. Code Snippet

```
def add_expense():

    name = name_entry.get()

    amount = amount_entry.get()



    if name == "" or amount == "":

        print("Please fill all fields")

    else:

        tree.insert("", "end", values=(name, amount))
```

6. Sample Test Cases

Input(Expense)	Output
Name = Food, Amount = 200, Category = Food, Date = 08-04-2026	Expense is added in table
Search = Food	Matching expense is selected
Select one expense and click Delete	Selected expense is removed
Amount field empty, Invalid date = 32-13-2026, Invalid amount = abc	Expense is not added

7. Challenges Faced

Designing a proper GUI layout using Tkinter required careful alignment of widgets.Managing button styling and color display on different operating systems (especially macOS) was slightly difficult and ensuring that the expense table updates correctly after each operation required testing.

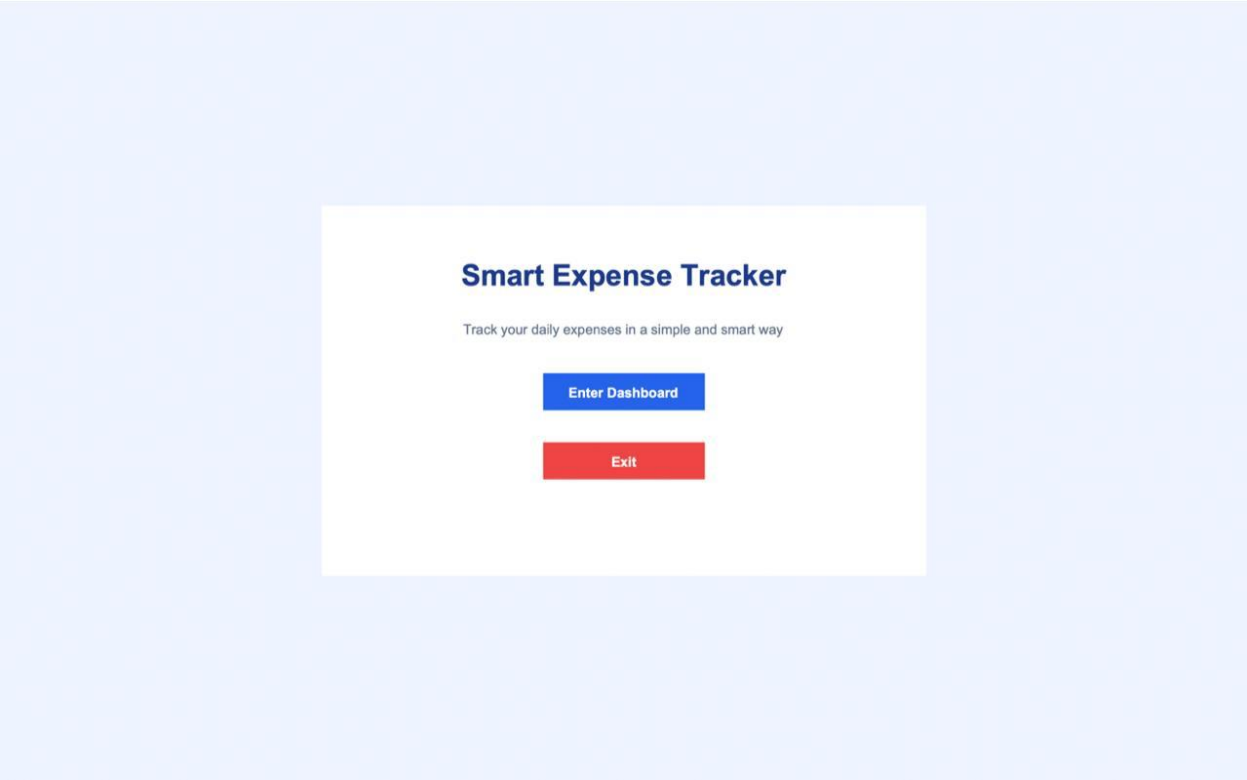
8. Results and Observations

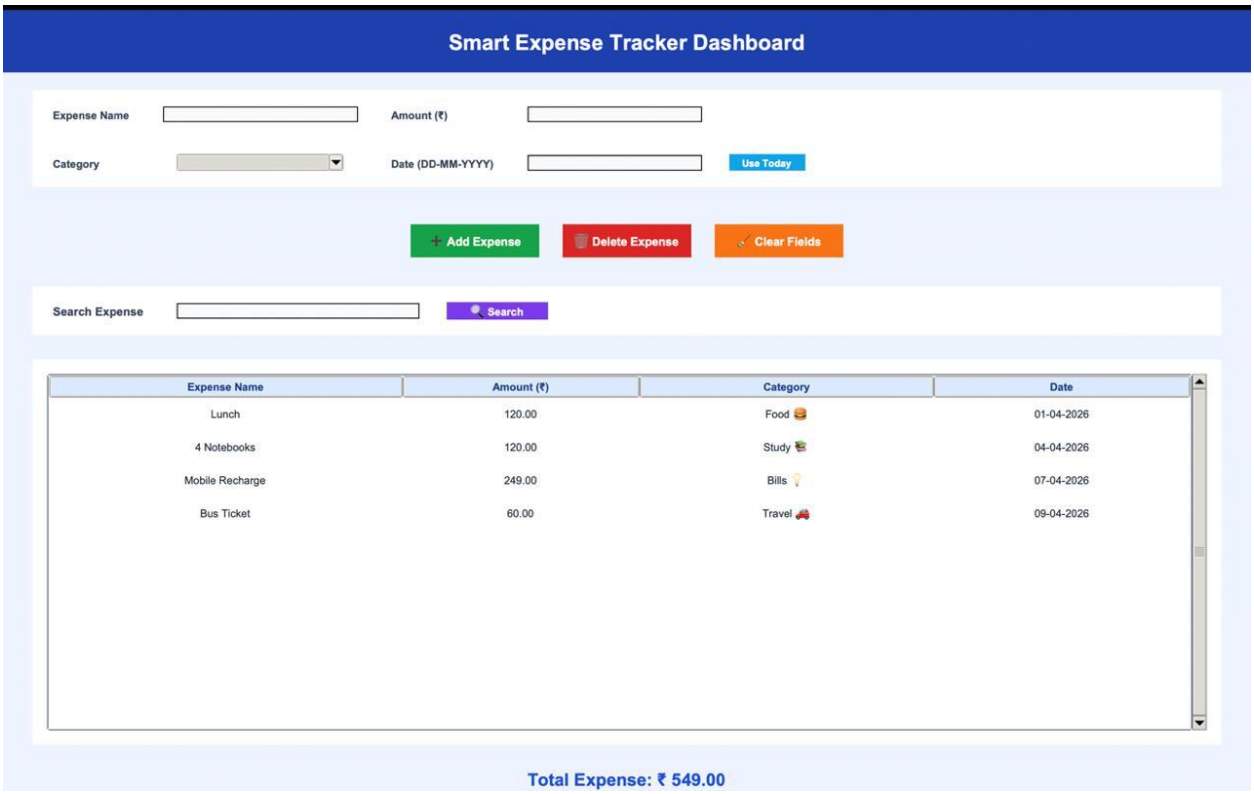
The Smart Expense Tracker project is working properly and all functionalities are implemented successfully. The system allows the user to add, delete, search, and manage expenses easily through a graphical interface.

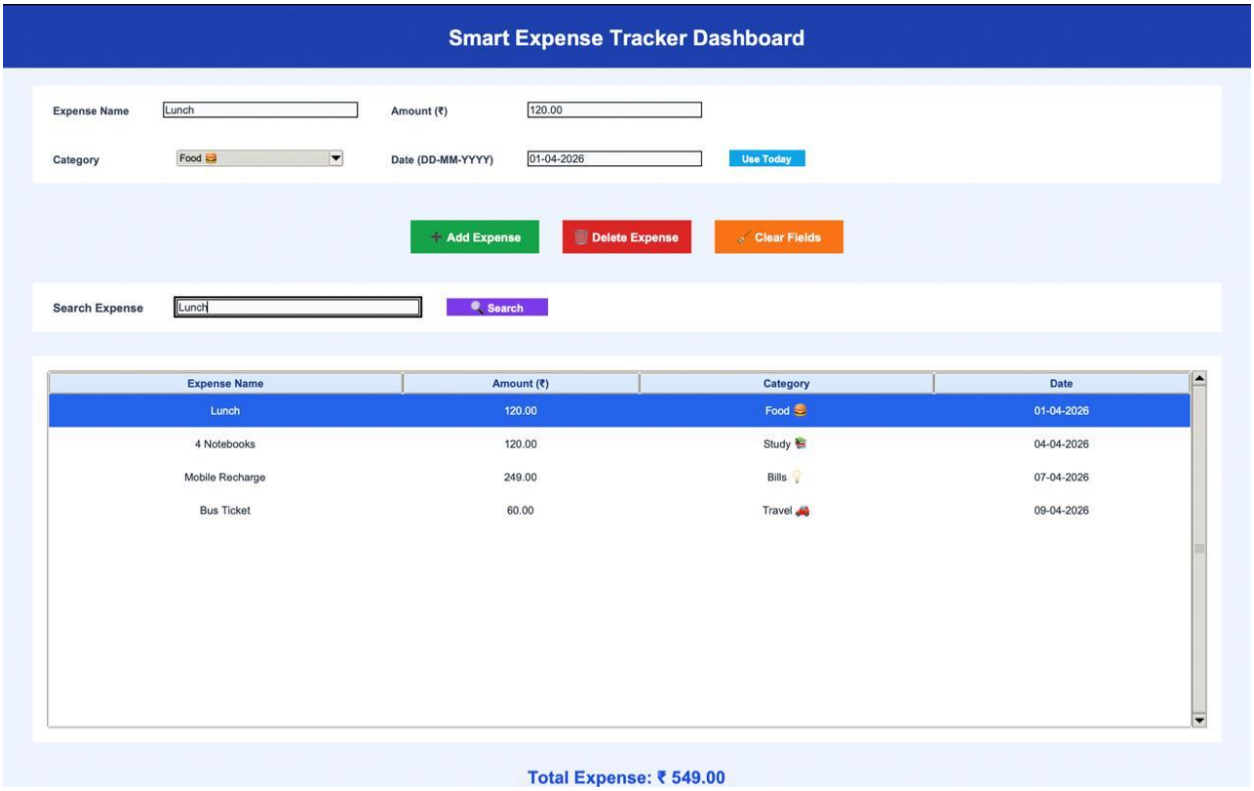
The entered data is validated and stored in a CSV file, and it is loaded correctly when the program starts. The total expense is also calculated automatically.

The screenshots of the welcome screen, adding expense, and searching expense are included to show the working of the project.

Overall, the project is simple, easy to use, and useful for managing daily expenses.




Smart Expense Tracker Dashboard
Expense Name
Amount (₹)
Category
Date (DD-MM-YYYY)
Use Today
Add Expense
Delete Expense
Clear Fields
Search Expense
Search
Expense Name
Amount (₹)
Category
Date
Lunch
120.00
Food
01-04-2026
4 Notebooks
120.00
Study
04-04-2026
Mobile Recharge
249.00
Bills
07-04-2026
Bus Ticket
60.00
Travel
09-04-2026
Total Expense: ₹ 549.00


Smart Expense Tracker Dashboard
Expense Name
Lunch
Amount (₹)
120.00
Category
Food
Date (DD-MM-YYYY)
01-04-2026
Use Today
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Search
Expense Name
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Category
Date
Lunch
120.00
Food
01-04-2026
4 Notebooks
120.00
Study
04-04-2026
Mobile Recharge
249.00
Bills
07-04-2026
Bus Ticket
60.00
Travel
09-04-2026
Total Expense: ₹ 549.00

Project Submission

Repository Link:
https://github.com/harshadabandichode/Expense-Tracker/blob/main/Expense_tracker

9. Signatures

Student Signature: _____

Guide Signature: _____

Date: _____